ECE301: Discrete-Time Signals & Systems

1. Nyquist-Shannon Sampling Theorem:

   The minimum sampling frequency fs must be at least twice the highest frequency fmax:

   fs >= 2 * fmax (Nyquist rate).

2. Discrete Fourier Transform (DFT):

   Transforms a sequence of N complex numbers into another sequence of frequency components.

3. Z-Transform and Region of Convergence (ROC):

   Analysis of Linear Time-Invariant (LTI) system stability depends on pole locations relative to unit ci